

MATHEMATICA Cheat Sheet: A8B17CAS Course

Useful Commands

Clear[x]	clears definitions of symbol x
ClearAll[x,y]	clears also Attributes of symbols x , y
Clear["mat*"]	clear all symbols whose name starts with "mat"
ClearAll["var*"]	clearall all symbols whose name starts with "var"
Quit[], Exit[]	terminates a Wolfram Language kernel session
ALT + .	terminate (long-running) computation
SHIFT + ENTER	evaluate current cell
%	previously evaluated result
%%	one before previous evaluated result
%%%	two before previous evaluated result
N[expr]	evaluate expr numerically

Built in Functions/Constants

Pi	constant π
Exp[]	exponential function
E	constant e
I	imaginary unit
Infinity	∞

Input forms of numbers

3*^6	3×10^6
2^^1001	$1001_2 = 9_{10}$, <i>base ^ digits</i>
123.456789`	machine-precision number (123.457)
123.456789`5	arbitrary precision number with precision 5 (123.46)
123.456789``5	arbitrary precision number with accuracy 5 (123.4568)

other number forms...¹.

MATHEMATICA uses proprietary format of numbers.

Types of Numbers, Tokens, Atomic Objects, and Related Commands

Integer	arbitrary-long integer
Real	approximate real number
Rational	rational number, numerator and denominator are Integers
Complex	complex number (with Real real and imaginary part)
Symbol	atomic object of type symbol
String	character string
SymbolName	extract name of a symbol
Characters	extract characters of a string
IntegerDigits	extract digits of an integer
RealDigits	extract digits of a real number
Numerator, Denominator	extract parts of a Rational number
Re, Im	extract parts of a complex number

Construction of Lists and Matrices

x = {1, 2, 3}	vector/list of 3 numbers; can be used as a row- and column-vector
x = {{1, 2}, {3, 4}}	matrix/(list of lists) 2 x 2, matrix = list of row-vectors
Range	form a list from a range of numbers
Table	create a (nested) list by a formula for each element
Array	a modification of Table
ConstantArray	a simplified Array for constant N-D arrays
CenterArray	places a given array to a center of a bigger one (with constant values)
Subdivide[a,b,n]	equivalent of MATLAB <code>linspace(a,b,n+1)</code>
IdentityMatrix[n]	equivalent of MATLAB <code>eye(n)</code>
DiagonalMatrix[vec,k]	equivalent of MATLAB <code>diag(vec,k)</code>

Related guides^{2,3}.

MATHEMATICA possible equivalents of other MATLAB functions

<code>zeros(n1,n2,...)</code>	Table[0,n1,n2,...]
<code>ones(n1,n2,...)</code>	ConstantArray[1,{n1,n2,...}]
<code>false(n1,n2,...)</code>	ConstantArray[False , {n1,n2,...}]

Calculus

D	partial derivatives of a vector functions
Integrate	symbolic integrals in one or more dimensions
ImplicitD	implicit derivatives
Grad	gradient of a scalar function
Div	divergence of a vector field
Curl	rotation of a vector field
Series	Taylor series (and generalized series)
Limit	a limit of a function
Sum, Product	a symbolic sum and product of a sequence

Related guide⁴.

Do not confuse **Sum** with **Total**.

Product of vector elements can be computed as *e.g.* **Times @@ vec**, resp. **Apply[Times, vec]**.

Sum of vector elements can be computed as *e.g.* **Total[vec]**, resp. **Plus @@ vec**, resp. **Apply[Plus, vec]**.

Expressions

FullForm	full-form of an expression (as seen by MATHEMATICA kernel)
TreeForm	tree-form of an expression
MatrixForm	matrix-form of an expression
Head	return Head of an expression
Length	gives number of elements/arguments of expression
Depth	returns depth of an expression/maximum number of indices to index in the expression
List	head of a list, {a,b,c} is equiv. to List[a,b,c]
Association	equivalent to Python dictionary
Hold	"hold" expression unevaluated
HoldForm	another variant of Hold
Evaluate	force evaluation of the expression
expr[[1,2,{3,4}]]	returns a subexpression of expr
Part[expr,1,2,{3,4}]	specified by indices
Span, ; ;	represent span of indices for indexing

Other forms of expression: **InputForm, OutputForm, StandardForm, TraditionalForm, TableForm**

¹<https://reference.wolfram.com/language/tutorial/InputSyntax.html#977>

²<https://reference.wolfram.com/language/guide/ConstructingLists.html>

³<https://reference.wolfram.com/language/guide/ConstructingMatrices.html>

⁴<https://reference.wolfram.com/language/guide/Calculus.html>

Cases	find occurrences of a patter in an expression
DeleteCases	inverse version of Cases
Select	version of Cases with a boolean function
Count	returns number of pattern maches in the given list
Rule, ->	a substitution rule
RuleDelayed, :>	a rule with delayed evaluation
Set, =	setting a variable
SetDelayed, :=	the Set with delayed evaluation
ReplaceAll, /. 	apply replacement rules to an expression
ReplaceRepeated, //. 	apply rules until the expression no longer changes
Sequence	repret an expression for a sequence of arguments

Scoping constructs

With	evaluate with a specified variables replaces by values
Module	localize names of variables and "hide" steps in a procedure
Function	a pure function with parameters #, resp. #1, #2,...
Function[u, 3+u], (3+#) &	example of pure functions, Function[u, 3+u][10] == 13 , (3+#) &[10] == 13

Patterns

x_, x: _	a pattern representing one expression, internally named as x
x__, x: __	a pattern representing a sequence of expressions, internally named as x
a b c	alternative pattern forms
Repeated, ..	repeat previous pattern
n_Integer, n: _Integer	pattern to match all expressions with head Integer , internally named as n
n_?IntegerQ	pattern to match all expression for witch IntegerQ returns true, internally named as n
n_/;IntegerQ[n] && n>3	a pattern to match all single expression for which the condition returns true

List Manipulation, and structural operations

Join	join lists/(expression with the same Head) at any level
Partition	split list/expression by a number of elements in sublist
Flatten	a reverse operation of Partition , flattens expression
FlattenAt	a version of Flatten
Transpose	transpose elements of a list/expression
Reverse	reverse order of elements in a list/expression
Sort	sorts arguments of the given expression
SortBy	version of Sort
Ordering	version of Sort that returns indices
Max	maximum of elements
MaximalBy	version of Max
Min	minimum of elements
MinimalBy	version of Min
Split	split expression by a condition
Gather	group arguments by a test
Riffle	generalized merging of lists/expression
ArrayReshape	equivalent of MATLAB <code>reshape(arr, n1, n2, ...)</code>

Applying functions

Map, /@	map a function/ wrap by the new head at any level of expression
Apply, @@	apply a function/change head at any level of expression
MapApply, @@@	apply to each element of list
MapIndexed	map with index notation
Nest	nesting function <i>n</i> -times
NestList	returns also a list of intermediate results
NestWhile	nesting function while a condition holds
NestWhileList	returns also a list of intermediate results
Fold	Fold function/symbol over a list of values
FoldList	also return intermediate results in a list
FixedPoint	repeatedly nest until a fixed point

Relational, boolean and related operators

Greater, >	greater than
GreaterEqual, >=	greater or equal than
Less, <	less than
LessEqual, <=	less or equal than
Unequal, !=	unequal

Test functions

NumericQ	true if the expr. is a numeric quantity
IntegerQ	true if integer
MemberQ	true if the <i>form/pattern</i> matches an element in a list
FreeQ	true if expr. is free of the <i>form</i> specified by a pattern
VectorQ	true if expr is a vector

Plot functions

Plot	simple 2D plot of a list of functions
LogPlot, LogLinearPlot, LogLogPlot	versions of Plot
Plot3D	3D plot
ContourPlot, DensityPlot	other plots to visualize 3D data
DiscretePlot	plot s sequence defined by a formula
DiscretePlot3D	plot s 2D sequence defined by a formula
ListPlot, ListLinePlot, ArrayPlot	plot lists of precomputed data
VectorPlot	plot a vector field

Some Plotting Options (valid only for certain plot-commands)

AspectRatio	ratio of height and width of figure
Filling	filling to insert under each curve
PlotLegends	legends of curves
PlotLabel	overall label of the plot
PlotRange	range of axes to display
PlotStyle	directive to specify style for each curve
AxesLabel	labels of axes
Frame	whether to draw a frame around figure
FrameLabel	equivalent to AxesLabel when Frame->True
GridLines	grid lines to draw
Ticks	axes ticks

Import, Export

Import	load data from a file/url to workspace
Export	export data from the workspace to a file